Mean Value Theorem (MVT)

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

provided $f(x)$ is differentiable in $a<x<b$

and continuous in $a \le x \le b$

$$\min f' \le \frac{f(b)-f(a)}{b-a} = f'(c) \le \max f' \quad (a<c<b)$$

$f'(c) > 0 \Rightarrow f(b) > f(a)$

$f'(c) < 0 \Rightarrow f(b) < f(a)$

$f'(c) = 0 \Rightarrow f(b) = f(a)$.

difference between MVT and linear approximation.

linear approximation: $\frac{\Delta f}{\Delta x} \approx f'(a)$ b near a ($b-a=\Delta x$)

MVT: $\frac{\Delta f}{\Delta x} = f'(c)$ some c between a and b

to calculate $\sqrt{102}$: $f(x)=\sqrt{x}$ $f'(x)=\frac{1}{2\sqrt{x}}$

linear approximation: $f(x) \approx f(a) + f'(a)(x-a)$

$\sqrt{102} \approx \sqrt{100} + f'(100)\cdot(102-100) = 10 + \frac{1}{10} = 10.1$

MVT: $f(x) = f(a) + f'(c)(x-a)$

$\sqrt{102} = f(100) + f'(c)\cdot(102-100) = 10 + \frac{1}{\sqrt{c}}$ $100<c<102$

inequality

① $e^x > x+1 \quad (x>0)$

$f(x) = e^x - (x+1) \quad f(0) = 1-1=0.$

$f'(x) = e^x - 1 > 0$ for $x>0$.

$f(x)$ is increasing from 0 so $f(x)>0$.

$\Rightarrow e^x > x+1.$

② $e^x > x+1+\frac{x^2}{2}$

$g(x) = e^x - (x+1+\frac{x^2}{2}) \quad g(0)=1-1=0.$

$g'(x) = e^x - (x+1) > 0$ from ①.

$g'(x) > 0 \Rightarrow g(x)$ is increasing from 0

$g(x) > 0 \Rightarrow e^x > x+1+\frac{x^2}{2}$

Differentials:

$y = f(x)$.

Differential of y: $dy = f'(x)\,dx \iff f'(x) = \frac{dy}{dx}$

use in linear approximation.

$$f(x) \approx f(a) + f'(a)(x-a).$$

$$f(x) - f(a) \approx f'(a)(x-a)$$

$$\Delta y \approx f'(a) \cdot \Delta x.$$

$$\Delta y \approx dy$$

use in MVT

$$\Delta y = f'(c)\Delta x \quad (c \in (a, a+\Delta x)).$$

~~Anti diff~~

Antiderivatives

← integral sign

$$\cancel{g(x)} = G(x) = \int g(x)\,dx.$$

antiderivative of g = indefinite integral of g

$$\int \sin x\,dx = -\cos x + C \qquad \int x^a\,dx = \frac{1}{a+1}x^{a+1} + C$$

$$a \neq -1$$

Method of substitution

example: $\int x^3(x^4+2)^5\,dx$

$u = x^4+2$

$\frac{du}{dx} = 4x^3$

$du = 4x^3\,dx$

$\int x^3(x^4+2)^5\,dx = \int \frac{u^5\,du}{4}$

$= \frac{u^6}{24} + C$

$= \frac{(x^4+2)^6}{24} + C$

advanced guessing

example: $\int e^{6x}\,dx$

$\int e^{6x}\,dx = \frac{e^{6x}}{6} + C$

guess e^{6x}

$\frac{d}{dx}(e^{6x}) = 6 \cdot e^{6x}$

① $\int \sin x \cos x\,dx = \frac{1}{2}\sin^2 x + C$

② $\int \sin x \cos x\,dx = -\frac{1}{2}\cos^2 x + C.$

$\Rightarrow$ actually they are the same

① − ② $= \frac{1}{2}(\sin^2 x + \cos^2 x) = \frac{1}{2}$

separation of variables

$$\frac{dy}{dx} = f(x)\, g(y)$$

$$\frac{dy}{g(y)} = f(x) \cdot dx$$

$$H(y) = \int \frac{dy}{g(y)} \qquad F(x) = \int f(x) \cdot dx.$$

$$H(y) = F(x) + C \text{ (implicit)}$$

$$y = H^{-1}(F(x) + C)$$

$$\frac{dy}{dx} = f(x) \Rightarrow dy = f(x)dx \Rightarrow \int dy = \int f(x)\, dx$$

example: $\left(\frac{d}{dx} + x\right) y = 0$

$$\frac{dy}{dx} + xy = 0. \qquad \ln|y| = -\frac{x^2}{2} + C.$$

$$\frac{dy}{dx} = -xy. \qquad e^{\ln|y|} = e^{-\frac{x^2}{2} + C}$$

$$\frac{dy}{y} = -x\,dx \qquad y = \pm A \cdot e^{-\frac{x^2}{2}} \quad A = e^{C}$$

$$\int \frac{dy}{y} = \int -x\,dx$$